

Exponential Family of Distributions

A probability density function gives the likelihood (probability) of a random variable to have a certain value. A probability curve plotted for a specific range of values of random variable is called a distribution e.g. normal distribution has a (inverted) bell-shaped curve. An important family of distributions is the exponential family.

The PDF of an exponential family of distributions is given by:

$$P(x | \eta) = h(x) \exp \{ \eta T(x) - A(\eta) \} \quad (1)$$

Here, η represents the natural parameter, $T(x)$ is the sufficient statistic, $h(x)$ is a normalizing constant, and $A(\eta)$ is the log partition function.

The distributions belonging to exponential family are Bernoulli, normal/Gaussian, negative exponential, poisson, gamma, categorical, etc.

For a Bernoulli distribution with $x \in \{0,1\}$ representing either success (1) or failure (0) of a trial and μ representing the probability of a success, $0 \leq \mu \leq 1$, we have

$$P(x | \mu) = \mu^x (1 - \mu)^{(1-x)}$$

$$P(x | \mu) = \exp \{ \log(\mu^x (1 - \mu)^{(1-x)}) \}$$

$$P(x | \mu) = \exp \{ x \log \mu + (1 - x) \log(1 - \mu) \}$$

$$P(x | \mu) = \exp \left\{ \left(\log \frac{\mu}{1 - \mu} \right) x + \log(1 - \mu) \right\}$$

Comparing the above expression with equation 1, we have,

$$\eta = \log \frac{\mu}{1 - \mu}$$

$$T(x) = x$$

$$h(x) = 1$$

$$A(\eta) = -\log(1 - \mu)$$

The most important distribution of exponential family is the normal/Gaussian distribution.

The probability density function of a normal distribution is:

$$P(x | \mu, \sigma^2) = \frac{1}{\sqrt{2\sigma^2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

where, μ is the mean of distribution, σ is standard deviation, σ^2 is the variance.

Generative Learners

A generative learner tries to learn the model that generates the data. It makes assumptions about the distributions.

Generative models contrast with discriminative models, in that a generative model is a full probabilistic model of all variables, whereas a discriminative model provides a model only for the target variable(s) conditional on the observed variables. Thus a generative model can be used, for example, to simulate (i.e. *generate*) values of any variable in the model, whereas a discriminative model allows only sampling of the target variables conditional on the observed quantities.

Examples of generative models are Gaussian/Linear Discriminant Analysis, Naïve Bayes and Hidden Markov Model.

Linear/Gaussian Discriminant Analysis

LDA makes some simplifying assumptions about your data:

- That your data is Gaussian, that each variable is shaped like a bell curve when plotted.
- That each attribute has the same variance, that values of each variable vary around the mean by the same amount on average.

With these assumptions, the LDA model estimates the mean and variance from your data for each class.

The discriminant function used by LDA is

$$f_i = \mu_i C^{-1} x_k^T - \frac{1}{2} \mu_i C^{-1} \mu_i^T + \ln(p_i)$$

Where C is the covariance matrix, μ_i is the mean of features in class i , x_k is data of row k , and p_i is the prior probability of a point to belong to class i .

We assign point/row k to class i that has maximum f_i .

Let the dataset be

X1	X2	Y
2.95	6.63	1
2.53	7.79	1
3.57	5.65	1
3.16	5.47	1
2.58	4.46	2
2.16	6.22	2
3.27	3.52	2

In the above given dataset, there are two classes – class 1 and class 2.

The prior probability of each class is $p_1=4/7$, $p_2=3/7$

The mean of features for each class is $\mu_1=[3.05 \ 6.38]$, $\mu_2=[2.67 \ 4.73]$

The mean of features of the whole dataset is $\mu=[2.88 \ 5.676]$

Now, we have to calculate the covariance between the feature vectors, for each class.

$$C = \begin{bmatrix} 0.206 & -0.233 \\ -0.233 & 1.689 \end{bmatrix}$$

Then we calculate the inverse covariance between the feature vectors, for each class.

$$C^{-1} = \begin{bmatrix} 5.745 & 0.791 \\ 0.791 & 0.701 \end{bmatrix}$$

For example, if we want to predict the class of point $[2.81 \ 5.46]$, then

$f_1=44.04$ and $f_2=44.08$

As $f_2 > f_1$, so the point is assigned class 2.